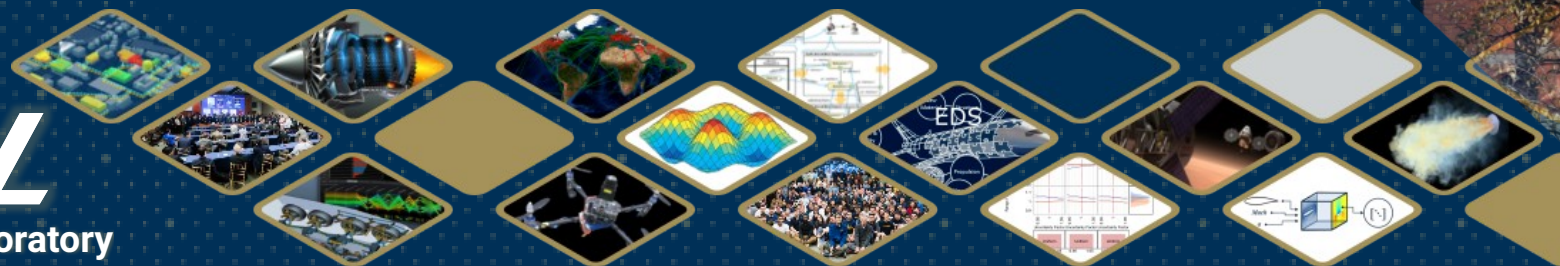
4699 SysML Team 102

Europa Submarine Model

Dhriti Naik, Anish Kulkarni, Ryland Marley, Aaditya Patel, Toby Reitman, William Mulder

July 22, 2026



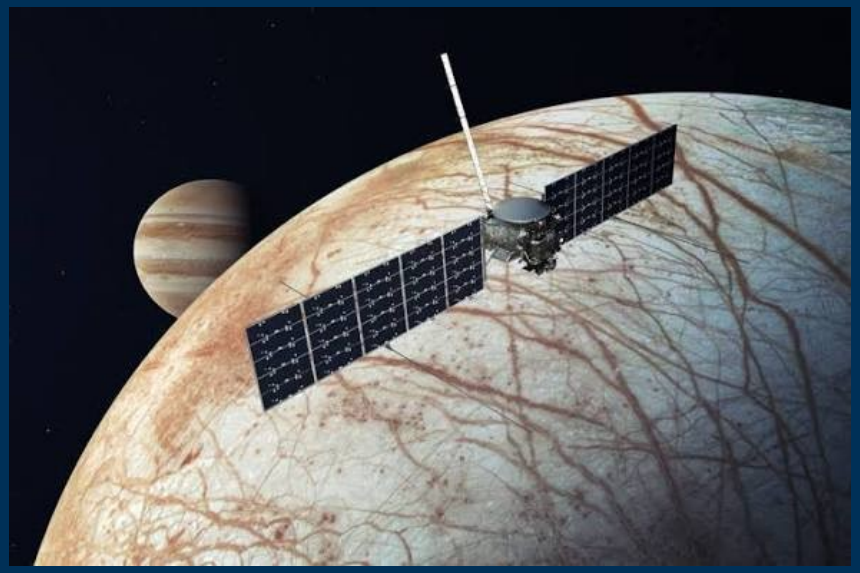


ASDL

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Background of Europa Exploration: Europa Clipper

- Europa is believed to have a saltwater ocean under its icy crust
- The chemical compounds and temperatures for liquid water to exist is thought to allow life to exist on Europa
- The Europa Clipper launched in 2024 and is intended to study the moon Europa that orbits Jupiter; a trip over 1.8 billion miles and five years
- It utilizes many systems to search for the necessary elements of life, such as water as it orbits the moon



Europa and Our Mission

- Until a few years ago, NASA was working on the Sensing With Independent Micro-swimmers (SWIM) mission
- Objective: use miniature submarines to explore under Europa's icy surface to search for signs of life
- Our project is modeling the subsystems of a SWIM submarine and the spacecraft transporting it, specifically Thermal Control, Power, Radiation Shielding and Structural, Communications, and Propulsion

NASA Prototypes >>



The Team



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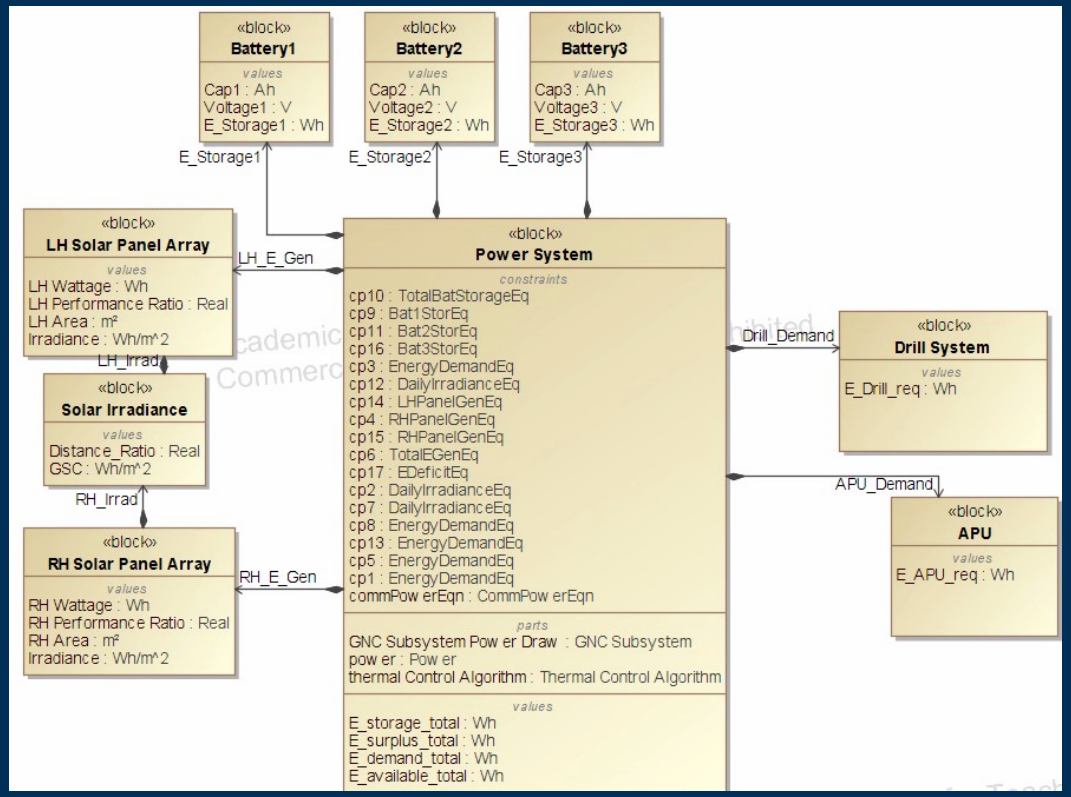
William Mulder
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Power Subsystem

William Mulder

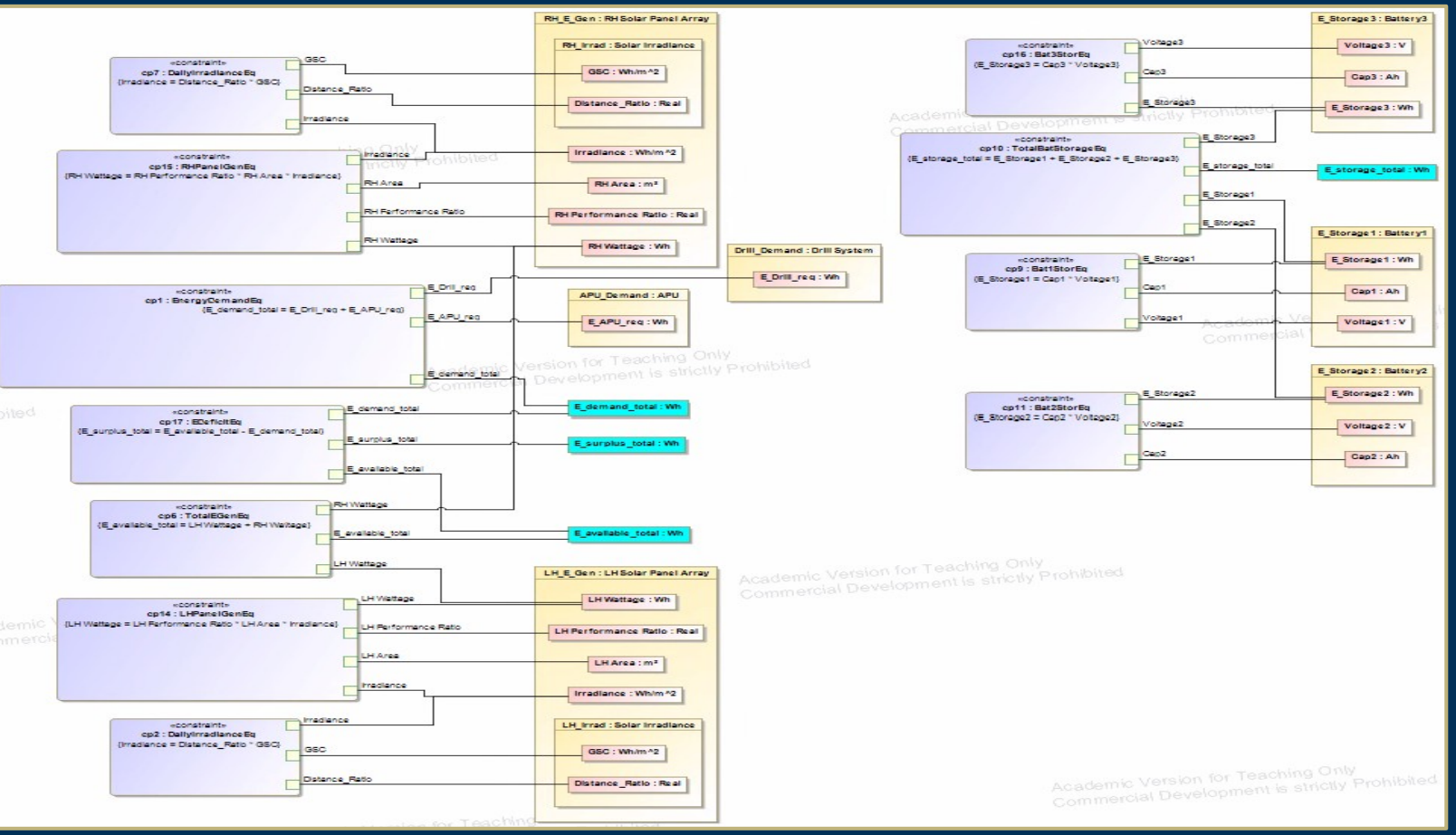
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Power Subsystem



- <u>OBJECTIVES:</u>
  - Calculate power generated by solar panels based on size and irradiance
  - Determine power storage for three batteries
  - Find power surplus/deficit from generation compared to demand from other systems.

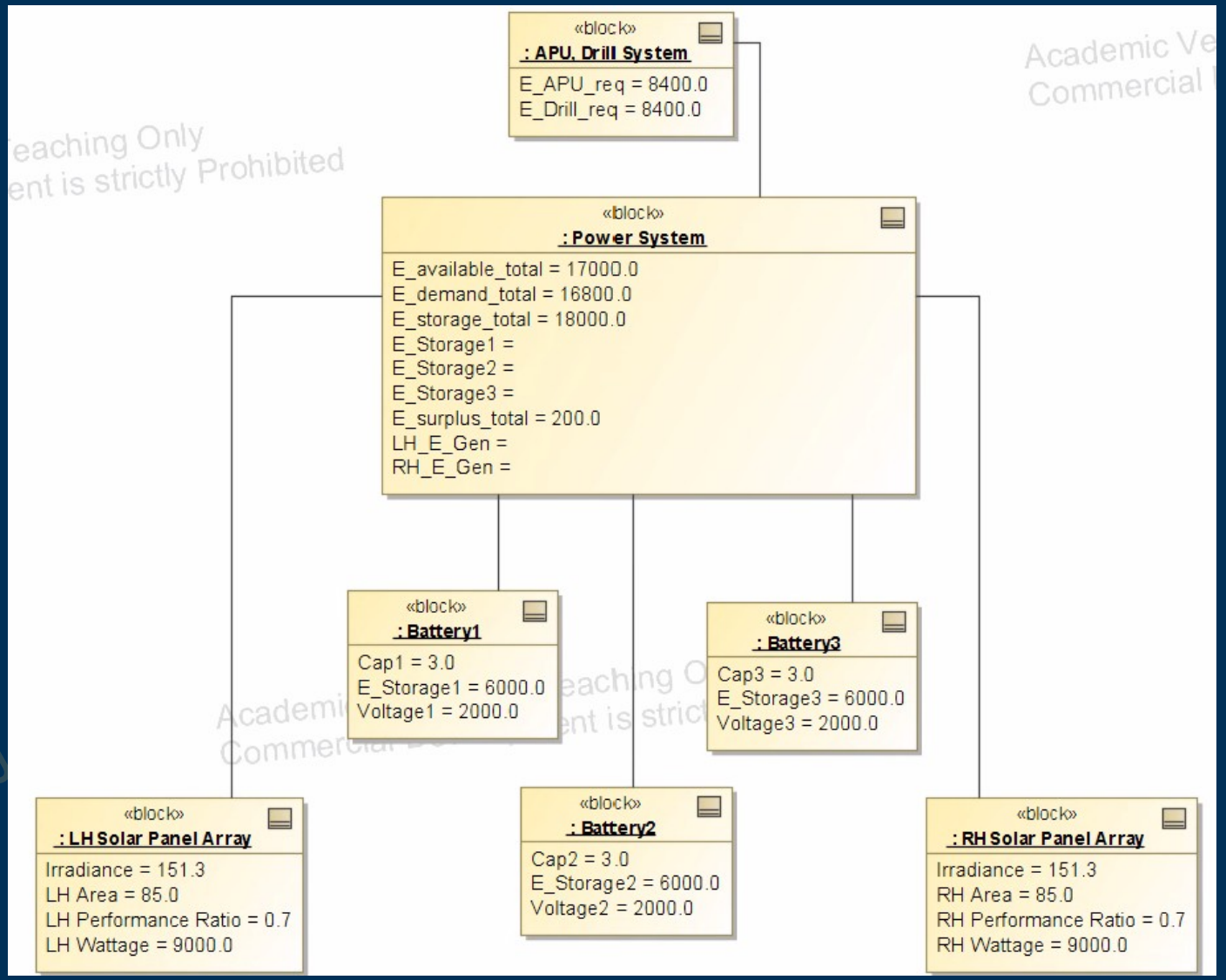
Power Subsystem Block Parametrics



- Assumptions
  - 100% efficiency from all systems except solar panels
  - Constant irradiance (GSC and distance ratio)
- Inputs
  - Solar panel wattage, performance ratio, area
  - Daily GSC and distance for irradiance
  - Battery capacity and voltage
  - Power consumption of other subsystems
- Equations: See footer for sources

1. https://science.nasa.gov/mission/europa-clipper/
2. https://apmonitor.com/dde/index.php/Main/EnergyStorage
3. https://www.bluettipower.com/blogs/articles/solar-power-calculation-formula
4. https://pvcase.com/blog/solar-irradiance-calculator

Power Example Instance



- Simple, but clearly highlights important connections and considerations for power system design
- Room for improvement in future semesters with development in simulation and test instances
- Further improvement in equation complexity



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Thermal Control Subsystem

Ryland Marley

Aerospace Systems Design Laboratory

Equations

- $Q_{solar} = S*A_p*\alpha_s*\cos(\theta)$
  - S - solar constant (distance dependent) (W/m^2)
  - A_p - projected area facing sun (m^2)
  - α_s - outer surface solar absorptivity
  - θ - incidence angle between rays and surface normal (simplified to 90)
- $Q_{albedo} = S*A_s*\alpha_s*a*F_{sat\text{-}planet}$
  - A_s - surface area receiving reflection (m^2)
  - a - planetary albedo coefficient
  - $F_{sat\text{-}planet}$ - view factor between satellite facet and planet
- $Q_{planet\text{-}IR} = q_{IR}*A_s*\varepsilon_{IR}*F_{sat\text{-}planet}$
  - q_{IR} - outgoing longwave radiation flux from planet (W/m^2)
  - ε_{IR} - infrared emissivity of satellites outer surface
- $Q_{internal} = P_{electrical}*(1-\eta)$
  - $P_{electrical}$ - total power consumed by internal electronics (W)
  - η - power efficiency of components
- $Qradiated = \varepsilon_{IR}*\sigma*A_r*T^4$
  - σ - Stefan-Boltzmann constant ($5.6704 \times 10^{-8}\ W/m^2 \cdot K^4$)
  - A_r - radiating surface area of satellite panels (m^2)
  - T - absolute temp of satellite surface (K)

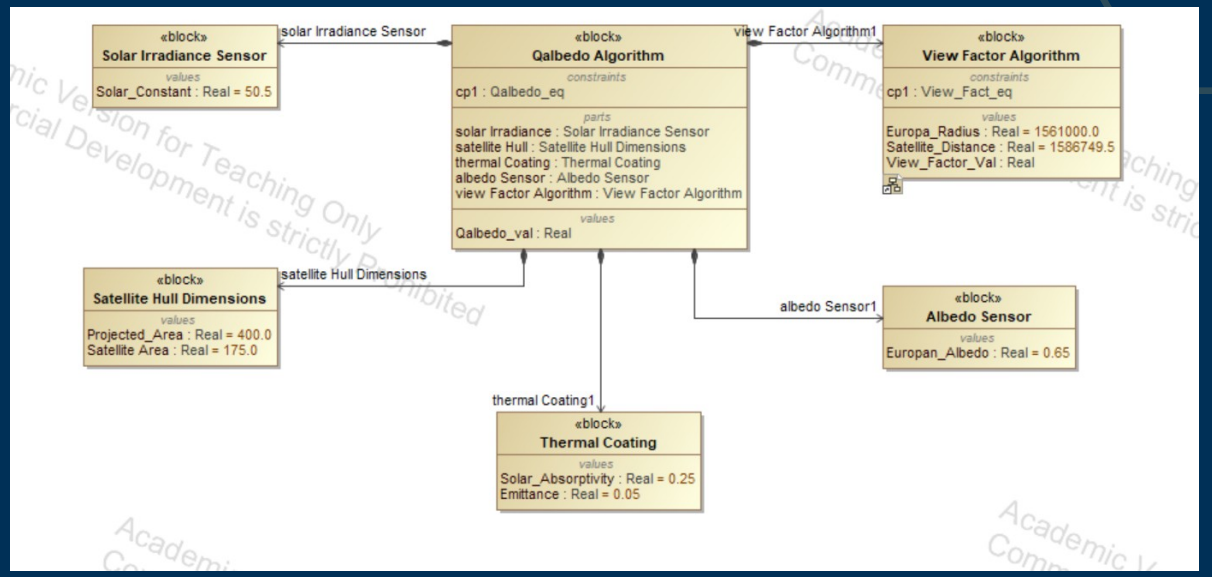
Governing Equation:

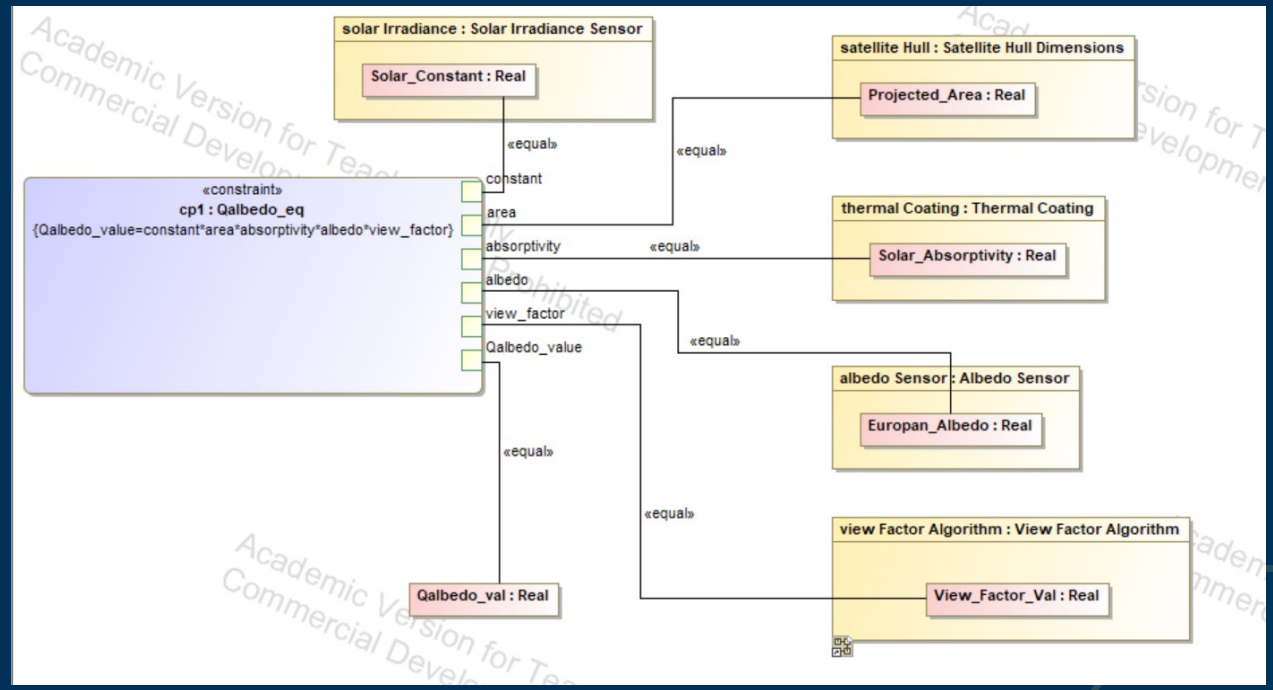
$$Q_{solar} + Q_{albedo} + Q_{planet\text{-}IR} + Q_{internal} - Q_{radiated} + Q_{heat} = 0$$

Albedo Parametrics

Assumptions:

- Near Europa
- Treating Satellite shape simply for area
- Orbiting at ~25,000m
- Gold plating (emittance, absorptivity)

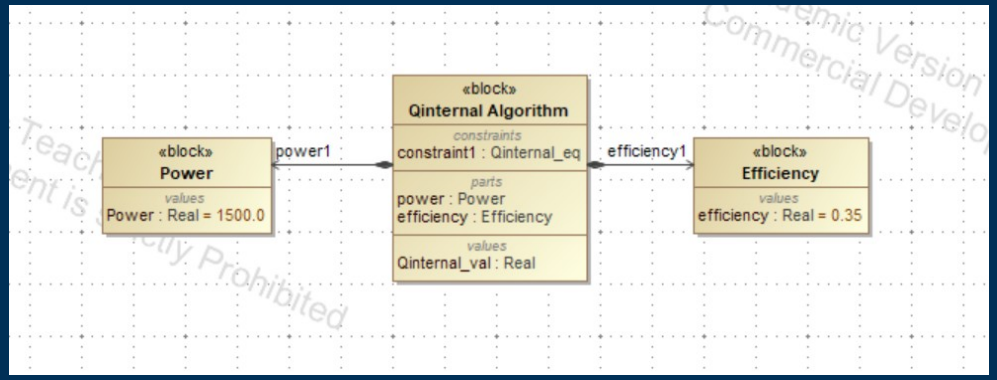


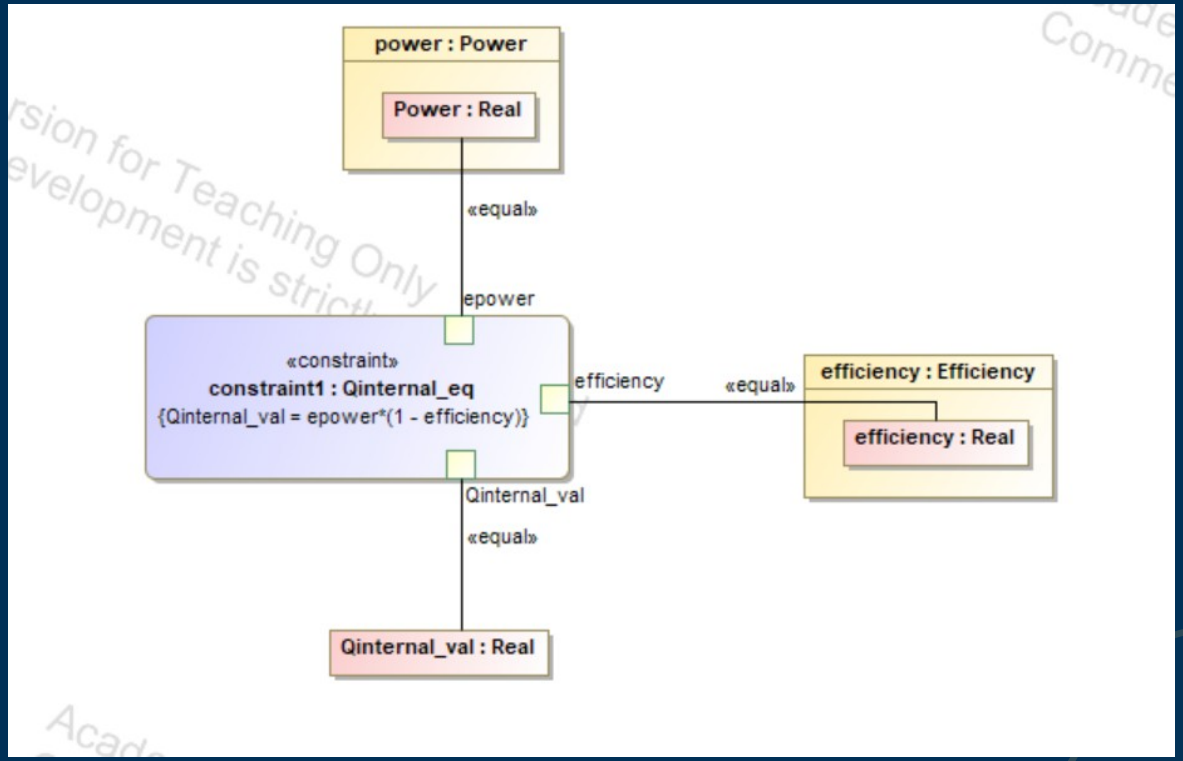


Internal Parametrics

Assumptions:

- Power (W)
- Efficiency similar to that of advanced NASA missions

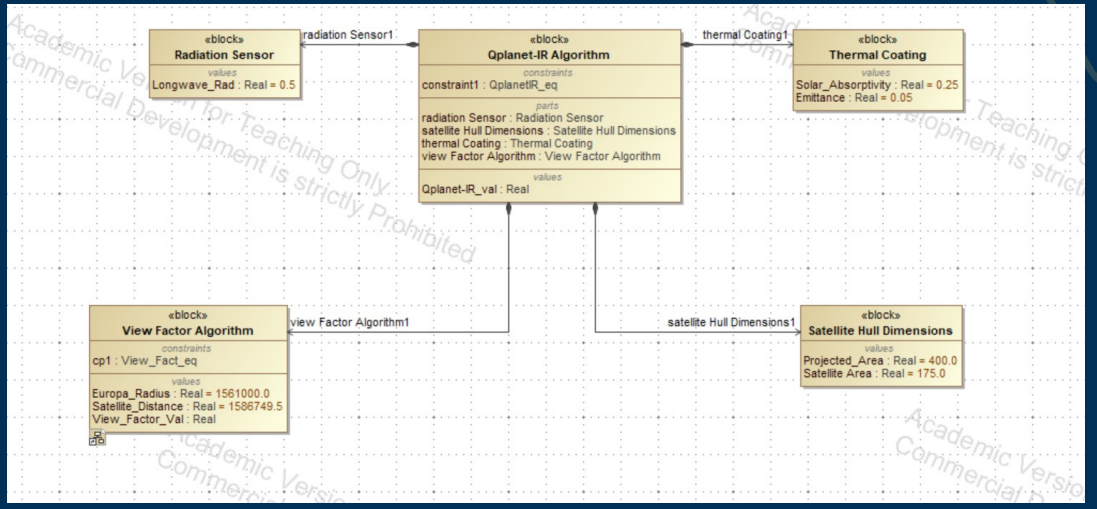


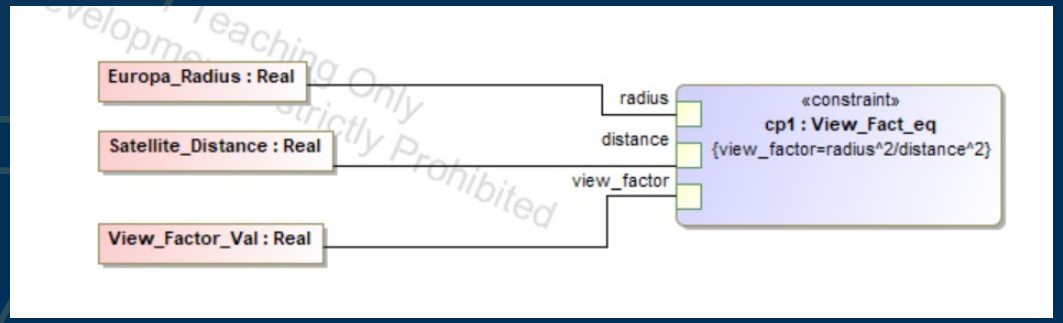


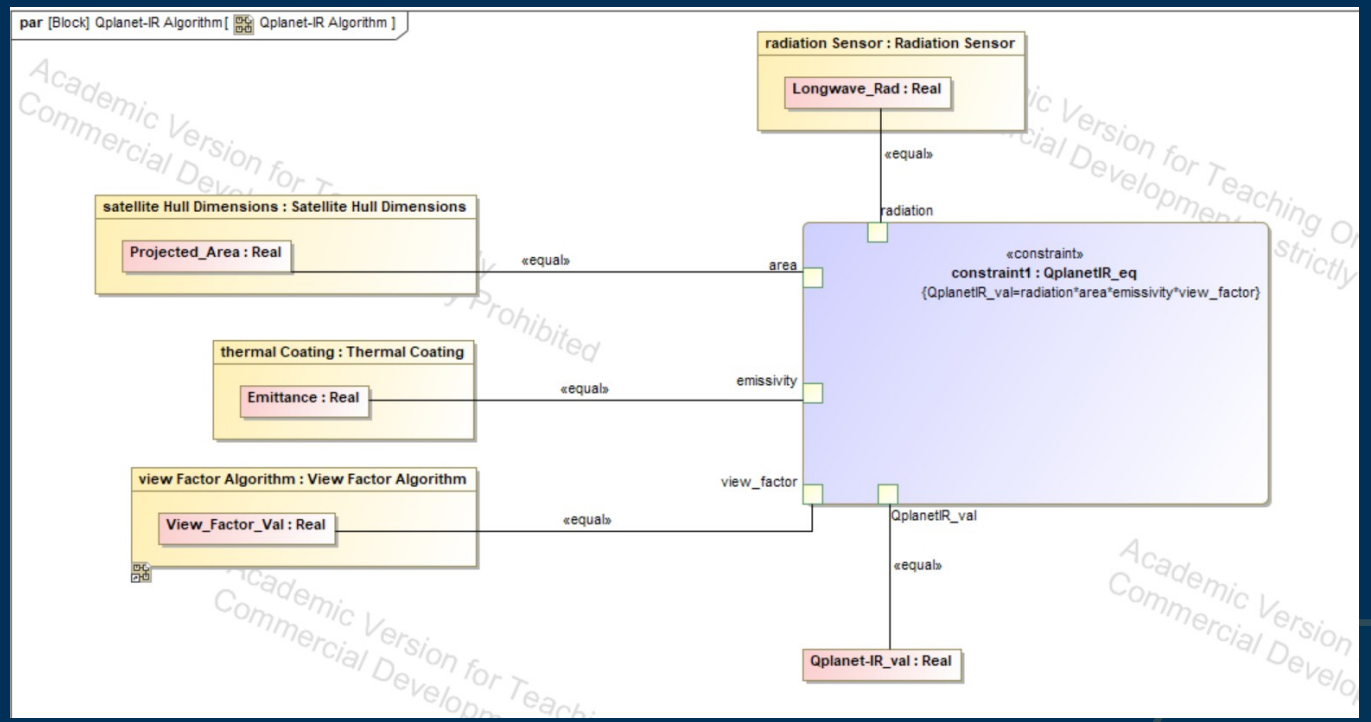
Planet-IR Parametrics

Assumptions:

- Longwave Radiation from values around Europan equator



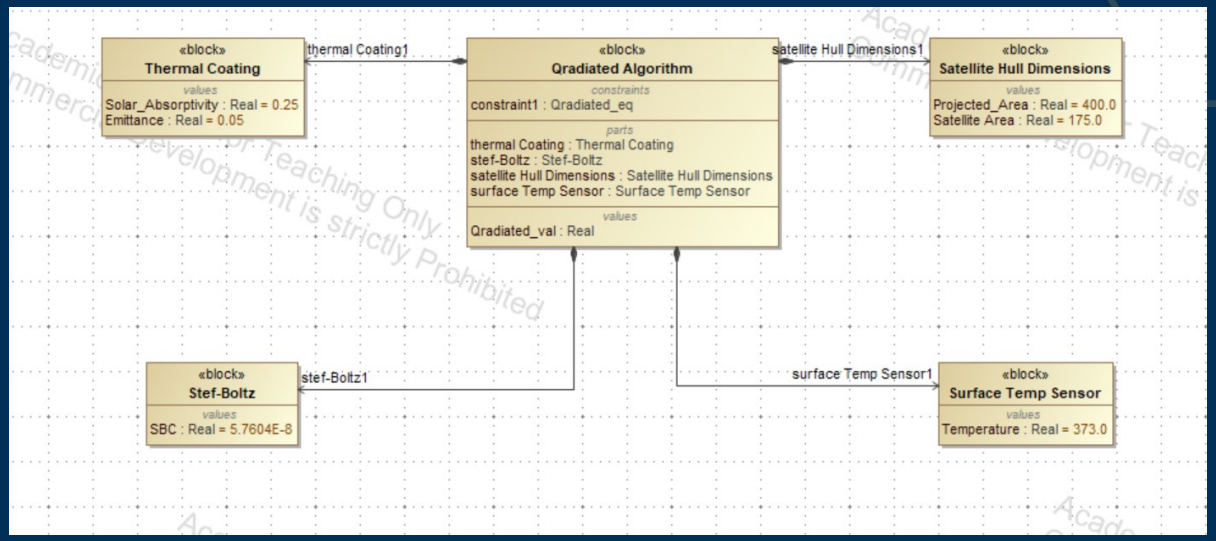


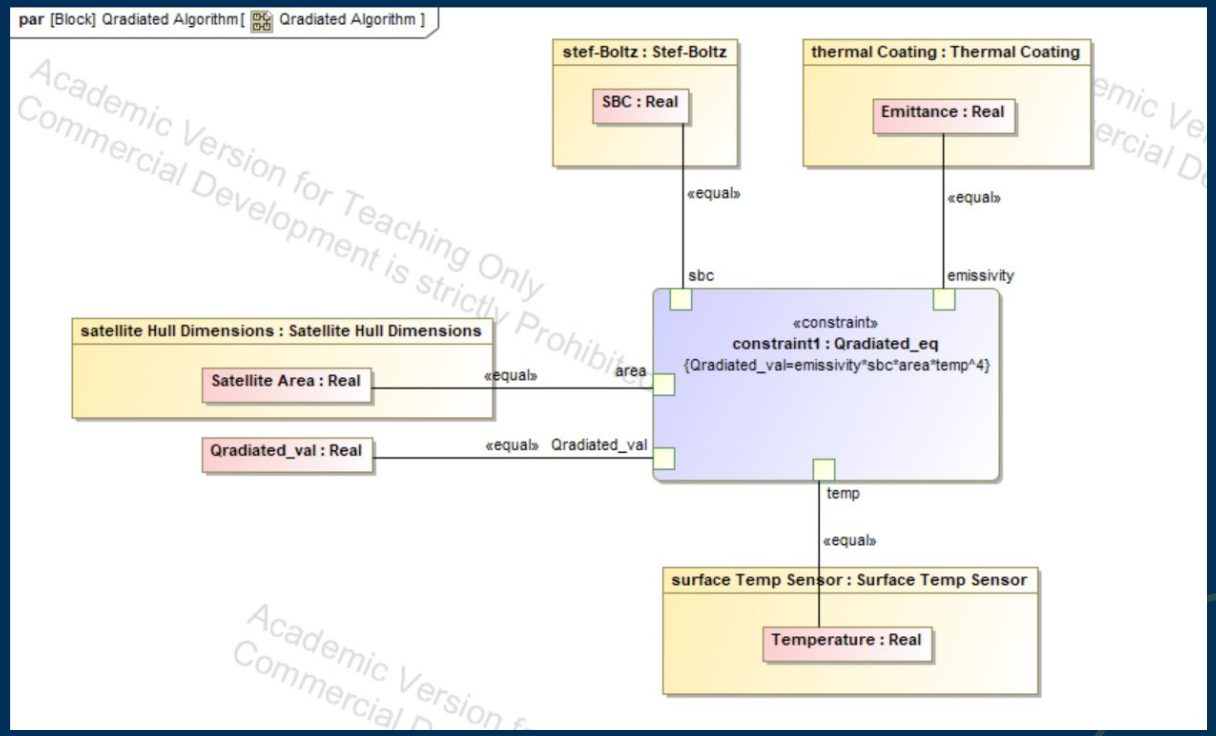


Radiated Parametrics

Assumptions:

- Surface is facing sun



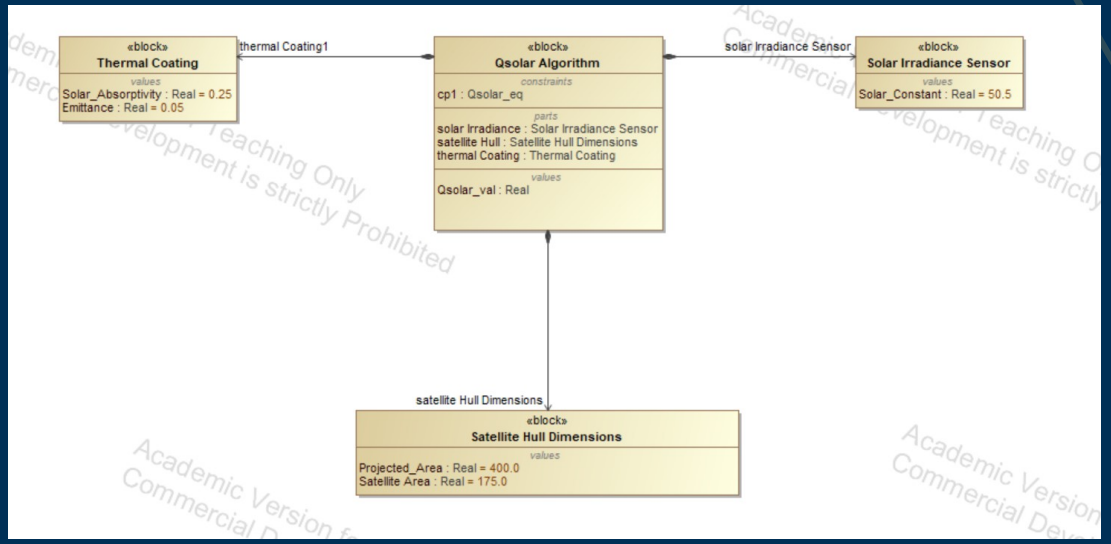


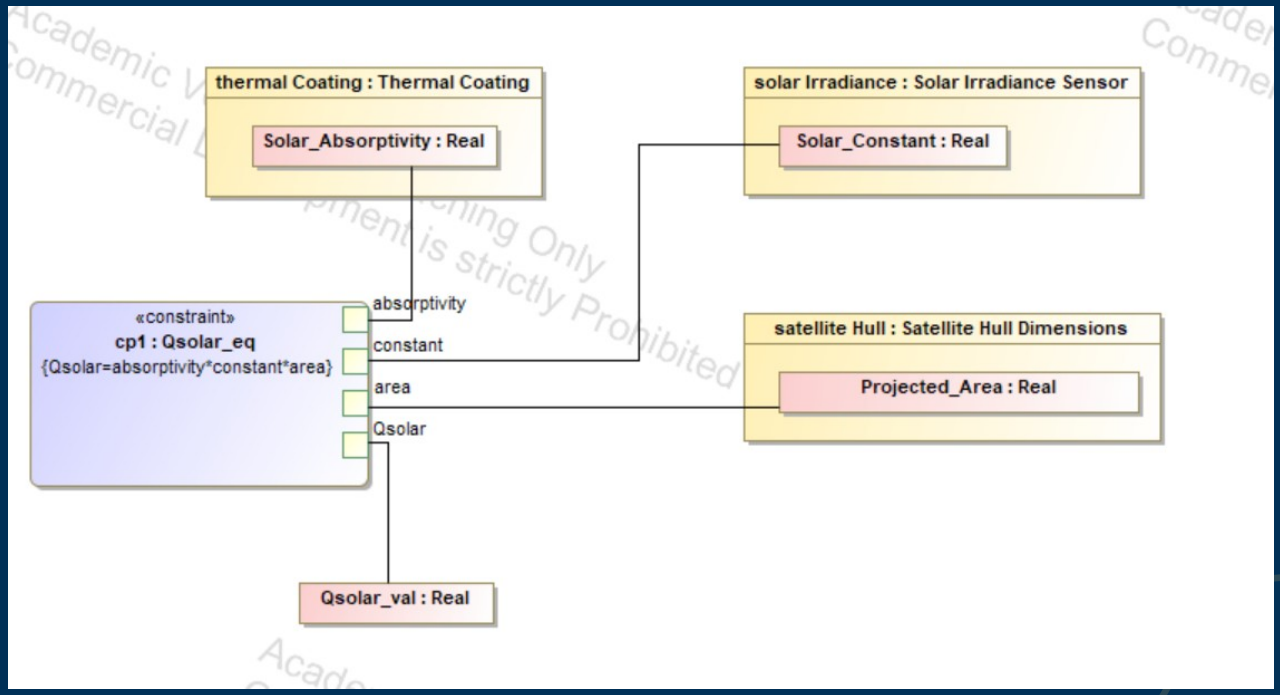
Academic Version for Teaching Only
Commercial Development is strictly Prohibited

Solar Parametrics

Assumptions:

- Solar constant value near Europa (W/m^2)

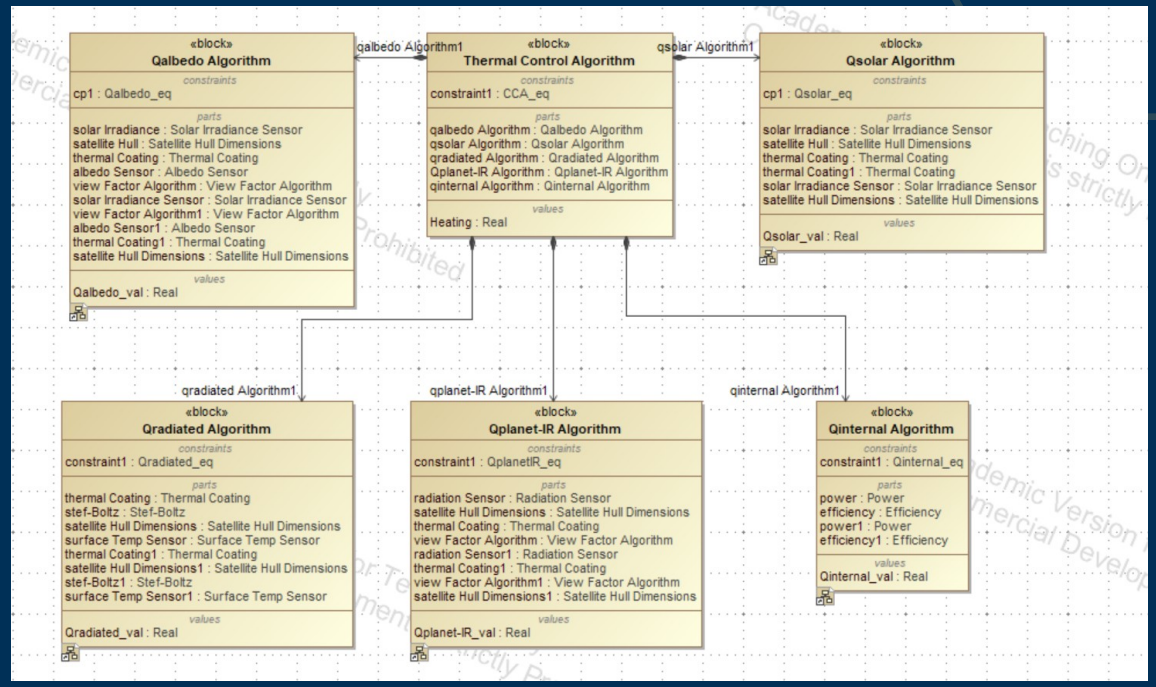


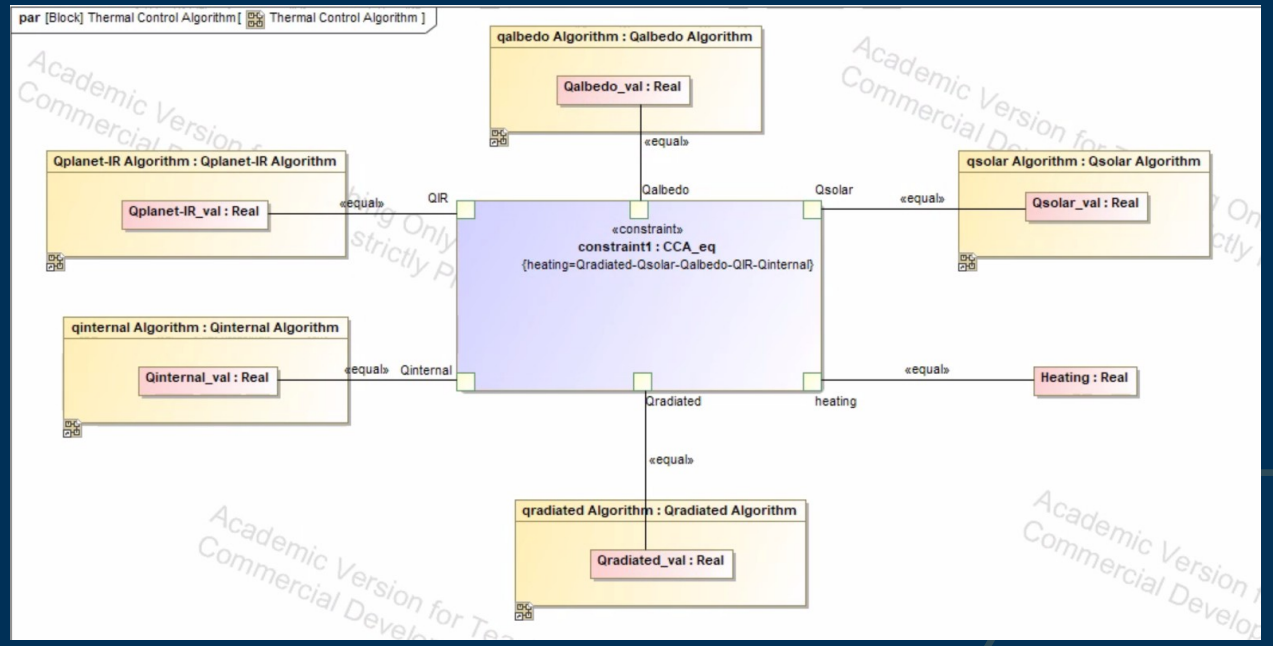


Heating Parametrics

- Calculations put together to determine how much power needed to maintain current temperature (W)

Thermal Control Algorithm	Thermal Control Algorithm@58935bd6
Heating : Real	545.0375
qalbedo Algorithm : Qalbedo Algorithm	Qalbedo Algorithm@526b30a6
Qalbedo_val : Real	3176.8287
qinternal Algorithm : Qinternal Algorithm	Qinternal Algorithm@3e406aea
Qinternal_val : Real	975.0000
Qplanet-IR Algorithm : Qplanet-IR Algorithm	Qplanet-IR Algorithm@3e43576f
Qplanet-IR_val : Real	9.6781
View_Factor_Val : Real	0.9678
qradiated Algorithm : Qradiated Algorithm	Qradiated Algorithm@763f0fc3
Qradiated_val : Real	9756.5443
qsolar Algorithm : Qsolar Algorithm	Qsolar Algorithm@48514413
Qsolar_val : Real	5050.0000





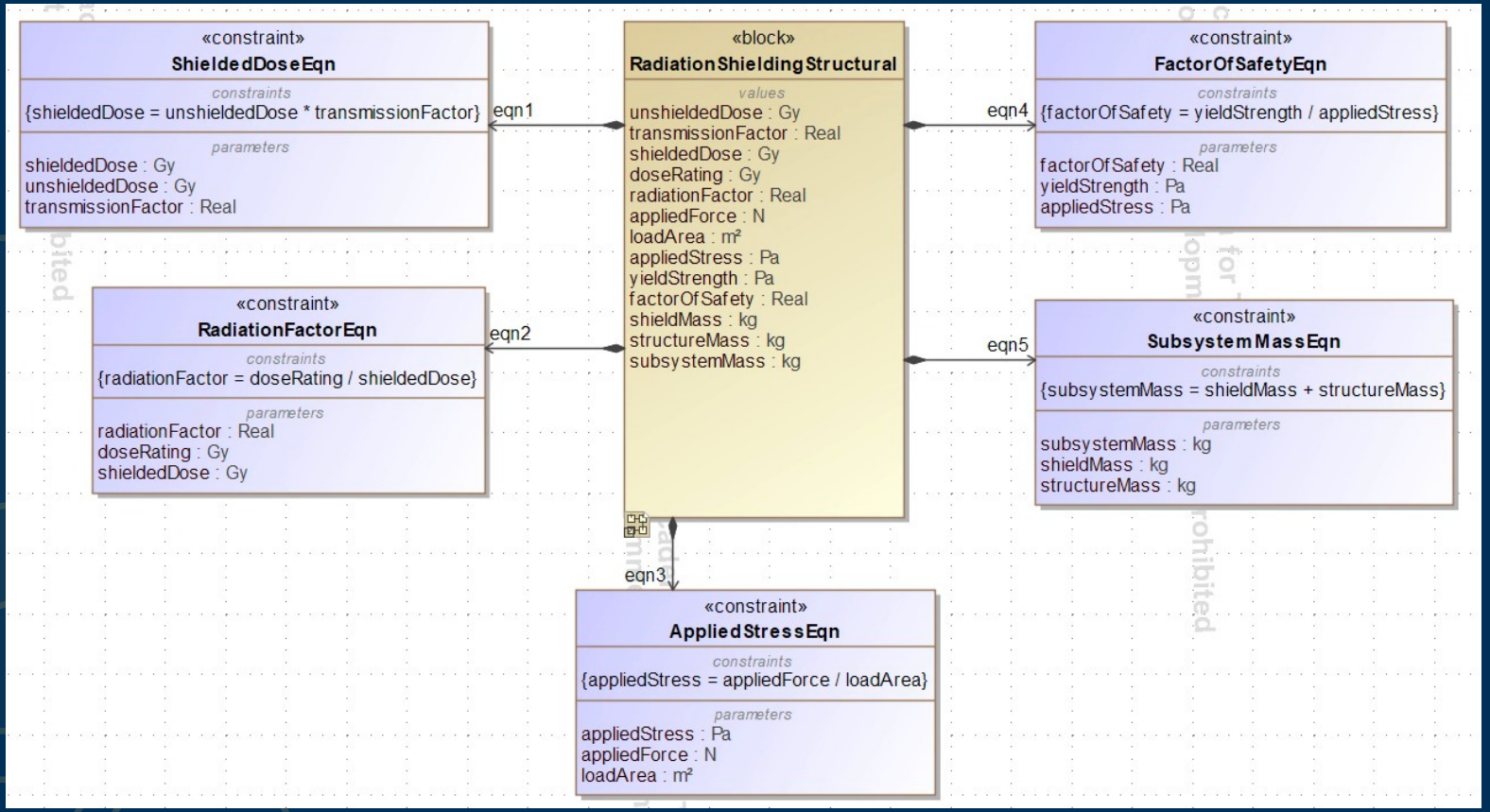
Radiation Shielding & Structural Subsystem

Aaditya Patel



Aerospace Systems Design Laboratory

Radiation Shielding & Structural Architecture



Objectives

- Reduce radiation exposure to electronics
- Evaluate structural stress and safety
- Calculate subsystem mass

Key outputs

- Shielded dose
- Radiation factor
- Applied stress
- Factor of safety
- Total subsystem mass

Assumptions

- Uniform structural loading
- Constant radiation transmission factor
- Preliminary conceptual analysis

Radiation and Structural Equations

Shielded radiation dose

- $D_{shielded} = D_{unshielded} T$
  - $D_{unshielded}$: dose before shielding
  - T : radiation transmission factor
  - $D_{shielded}$: dose reaching protected electronics

Radiation factor

- $RF = \frac{D_{rating}}{D_{shielded}}$
  - D_{rating} : qualified electronics dose rating
  - $RF > 1$: rating exceeds predicted shielded dose

Applied Stress

- $\sigma = \frac{F}{A}$
  - F : applied force
  - A : load-bearing area

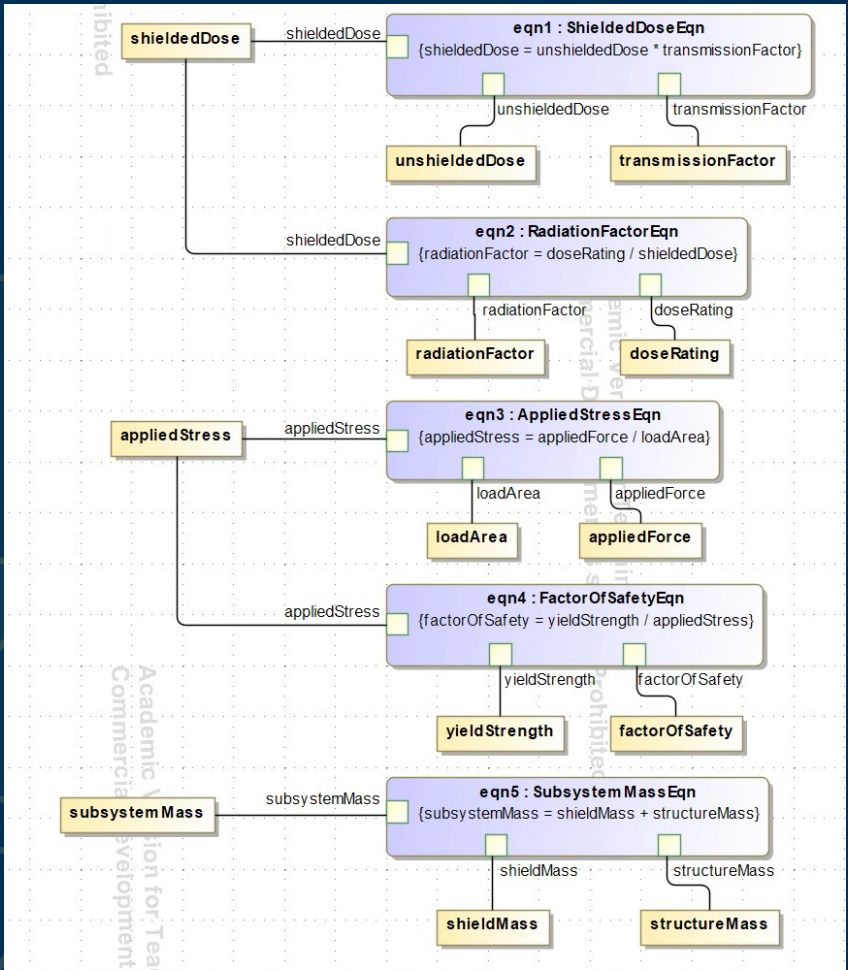
Structural factor of safety

- $FS = \frac{\sigma_y}{\sigma}$
  - σ_y : material yield strength
  - $FS > 1$: predicted stress is below yield

Subsystem Mass

- $m_{subsystem} = m_{shield} + m_{structure}$

Radiation Shielding & Structural Parametrics



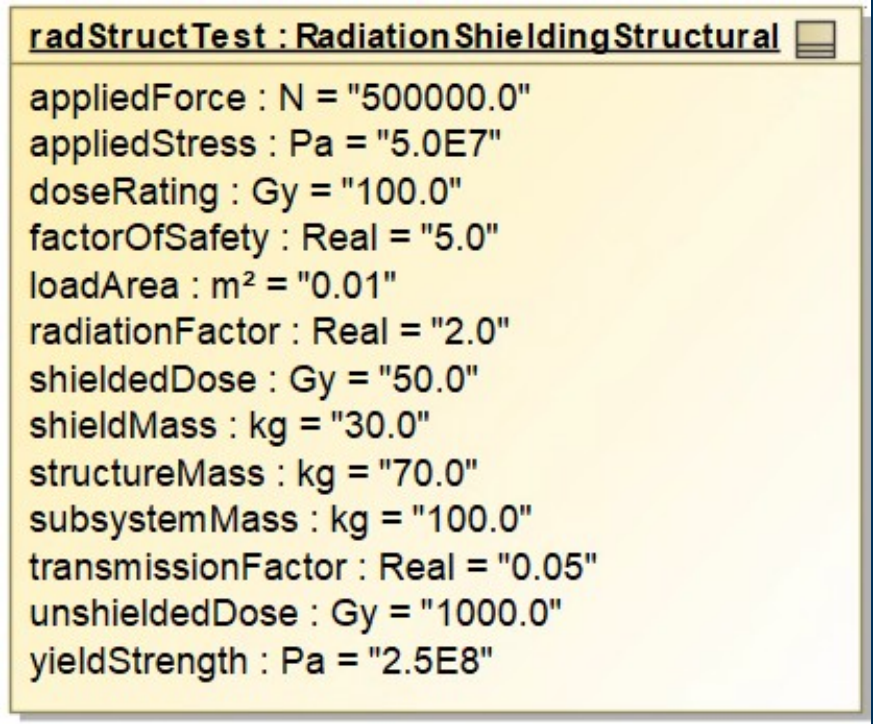
Calculation flow

- Shielded dose is calculated from unshielded dose and transmission factor.
- Structural stress is calculated from applied force and load area.
- Radiation factor, factor of safety, and subsystem mass are then determined.

Scope

- The model assumes constant radiation transmission and uniform loading.

Radiation & Structural Example Instance



Key results

- Shielding reduced the modeled dose from **1,000 Gy to 50 Gy**, a **95% reduction**.
- The electronics rating was **twice the calculated shielded dose**.
- The structural case produced **stress of 50 MPa** and a **factor of safety of 5**.
- Shielding and structure contributed **100 kg** to the total vehicle mass.

Values shown are for model verification and are not final Europa design inputs.



Communication Subsystem

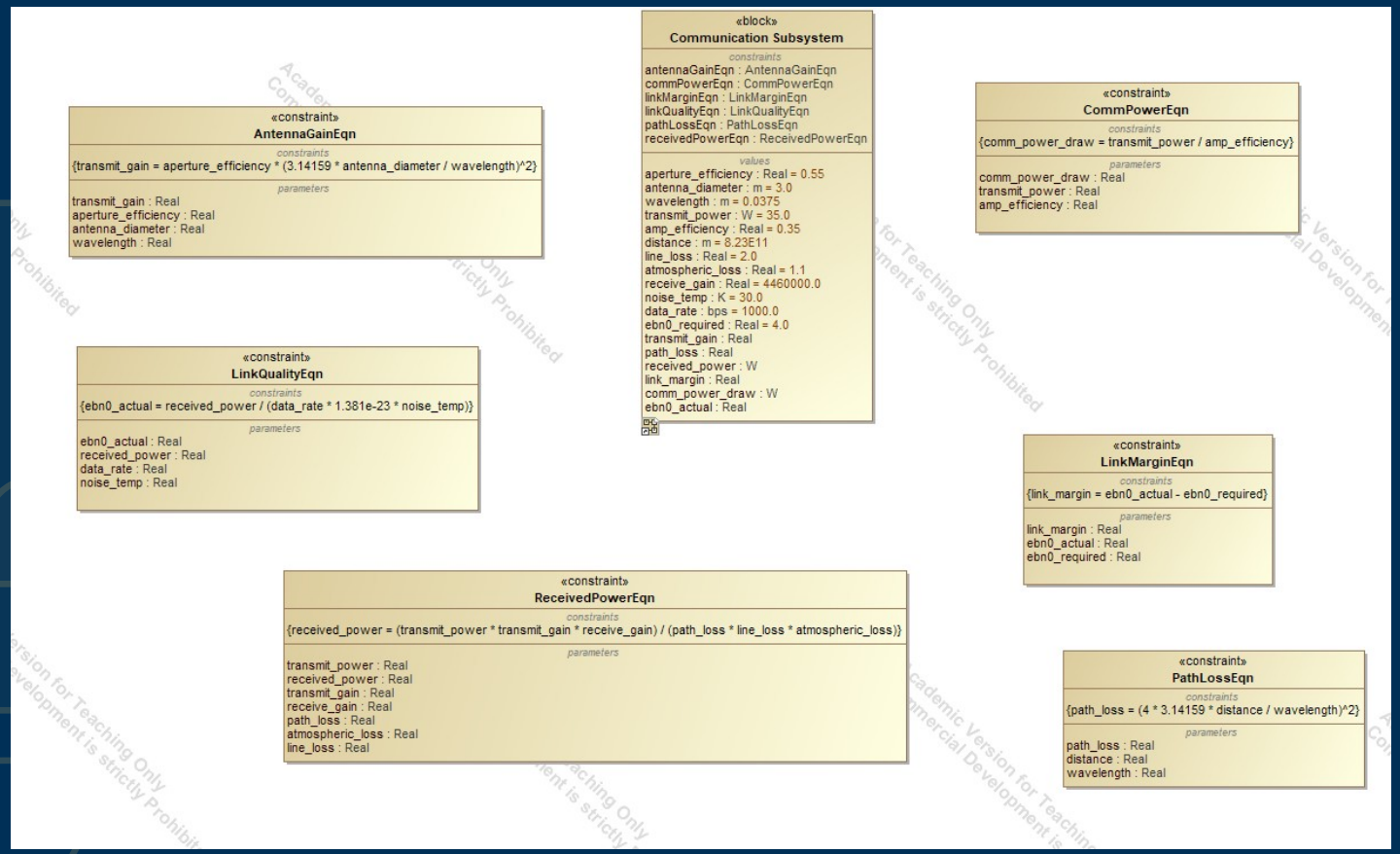
Anish Kulkarni
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By Anish Kulkarni



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Communication System Architecture

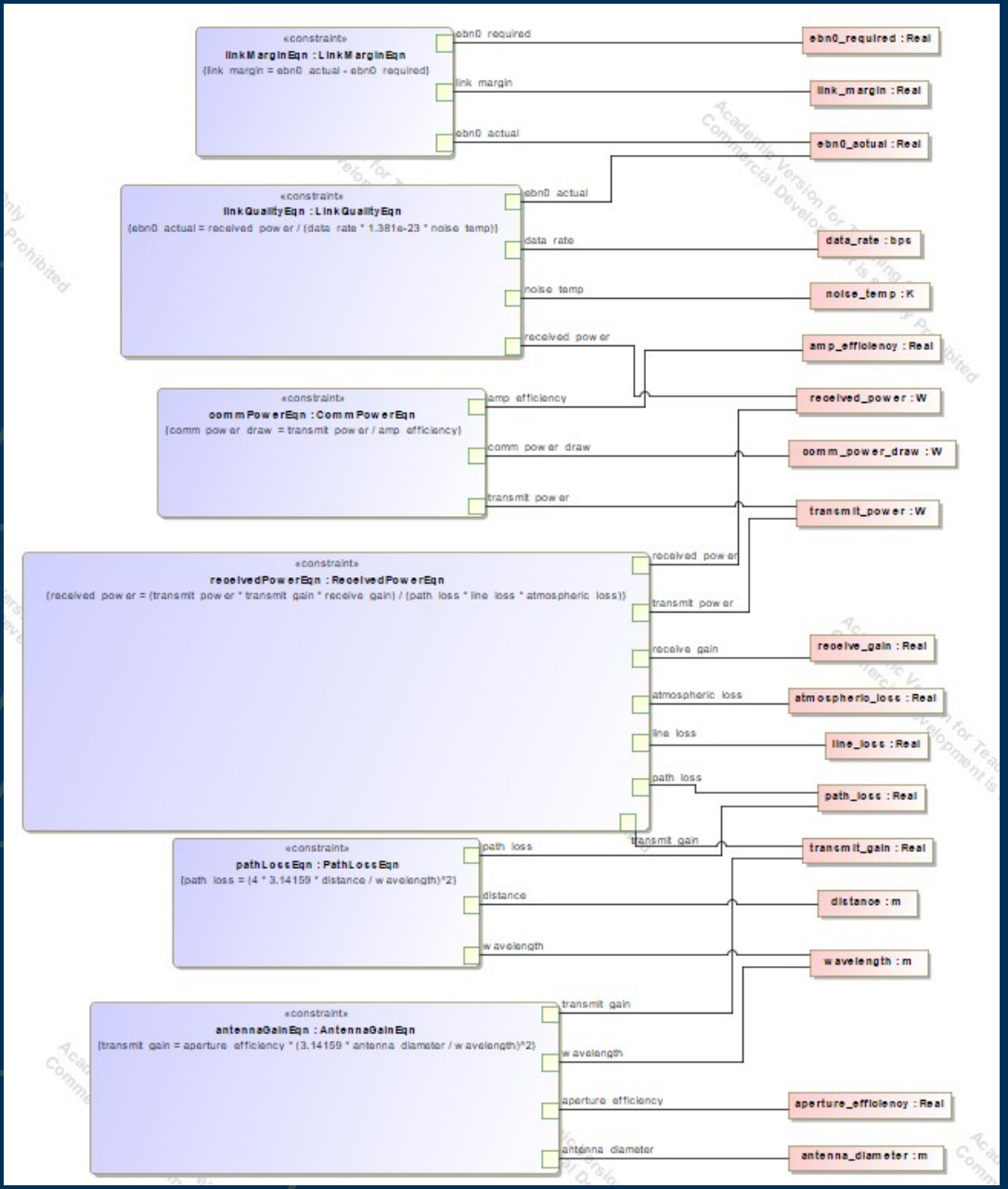


- OBJECTIVES:
  - Close the RF link from the Europa lander to Earth across 5 AU
  - Verify enough signal arrives to decode science data reliably
  - Report the radio's power draw to the power subsystem
- Key Outputs
  - Transmit antenna gain
  - Free space path loss
  - Received power at the DSN
  - Energy per bit to noise density (Eb/N0)
  - Link margin
  - Comm power draw and daily energy demand

Communication Equations Used

- Antenna Gain: $G = \eta(\pi D/\lambda)^2$
  - η = aperture efficiency
  - D = antenna diameter
  - λ = wavelength ($\lambda = c/f$)
- Free-Space Path Loss: $L_{fs} = (4\pi d/\lambda)^2$
  - d = range, Europa to Earth
  - λ = wavelength
- Received Power: $P_r = (P_t \cdot G_t \cdot G_r)/(L_{fs} \cdot L_l \cdot L_a)$
  - P_t = transmit power
  - G_t, G_r = spacecraft and DSN antenna gain
  - L_l = line and pointing loss
  - L_a = atmospheric loss at receiver
- Link Quality: $E_b/N_0 = P_r/(R \cdot k \cdot T_s)$
  - R = data rate
  - k = Boltzmann constant
  - T_s = system noise temperature
- Link Margin: $M = (E_b/N_0)_{actual} - (E_b/N_0)_{required}$
  - $M > 0$: link closes with margin to spare
- Comm Power Draw: $P_{comm} = P_t/\eta_{pa}$
  - η_{pa} = power amplifier efficiency

Communication System Architecture



- Assumptions:
  - Free space propagation with fixed pointing and atmospheric loss factors
  - Constant data rate during a downlink pass
  - Range from GNC is closest approach so margin is optimistic
- Inputs:
  - Antenna diameter, apperture efficiency, wavelength
  - Transmit power and amplifier efficiency
  - DSN receive gain and system noise temperature
  - Data rate and required Eb/N0
  - Range from GNC
- Calculation flow:
  - Antenna gain and path loss set received power
  - Received power and data rate set Eb/N0
  - Eb/N0 against the requirement gives link margin, the pass/fail metric

Communication Sample values used for calculation

ParaMagic(R) 18.0 - communication Subsystem

Name	Qualified Name	Type	Causality	Values
Communication Subsystem	Communication Subsystem1::communication Su...	Communication Subsystem		
amp_efficiency		Real	given	0.350
antenna_diameter		m	given	3.000
aperture_efficiency		Real	given	0.550
atmospheric_loss		Real	given	1.100
comm_power_draw		W	target	100.000
data_rate		bps	given	1,000.000
distance		m	given	823,000,000,000.000
ebn0_actual		Real	target	78.226
ebn0_required		Real	given	4.000
line_loss		Real	given	2.000
link_margin		Real	target	74.226
noise_temp		K	given	30.000
path_loss		Real	target	76,059,966,421,745,730,...
receive_gain		Real	given	4,460,000.000
received_power		W	target	0.000
transmit_gain		Real	target	34,740.949
transmit_power		W	given	35.000
wavelength		m	given	0.037

- Solved from antenna geometry to margin
- Margin is positive so the link closes with the given hardware

1. https://deepspace.jpl.nasa.gov/dsndocs/810-005/
2. https://science.nasa.gov/missions/europa-clipper/europa-clipper-resources/standing-tall-europa-clippers-high-gain-antenna/
3. https://www.nasa.gov/centers-and-facilities/langley/europa-clipper-high-gain-antenna-undergoes-precision-testing-at-nasa-langley/
4. https://link.springer.com/article/10.1007/s11214-023-00972-0
5. https://ieeexplore.ieee.org/abstract/document/8058576/



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Propulsion Subsystem

Dhriti Naik

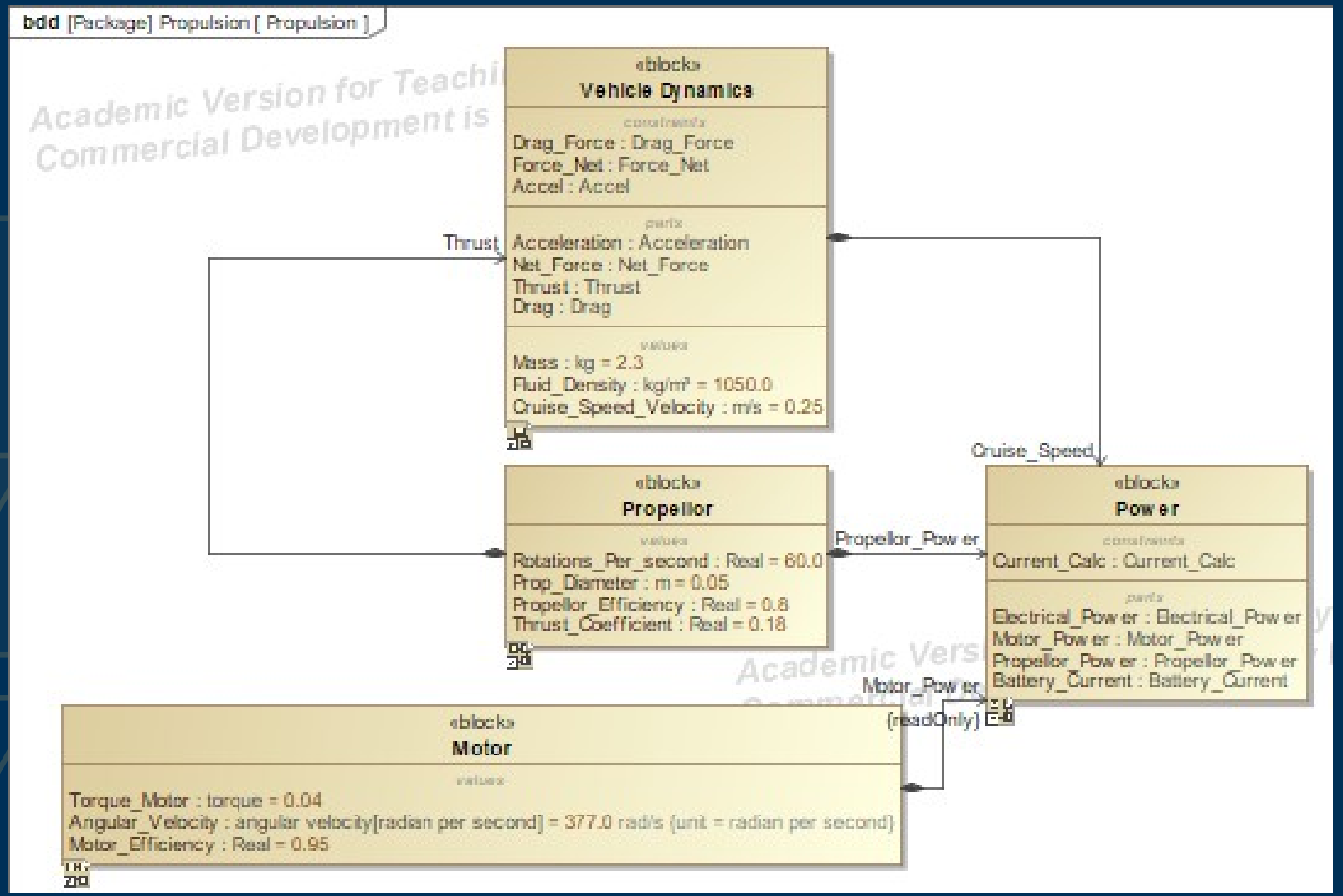


Aerospace Systems Design Laboratory

Equations

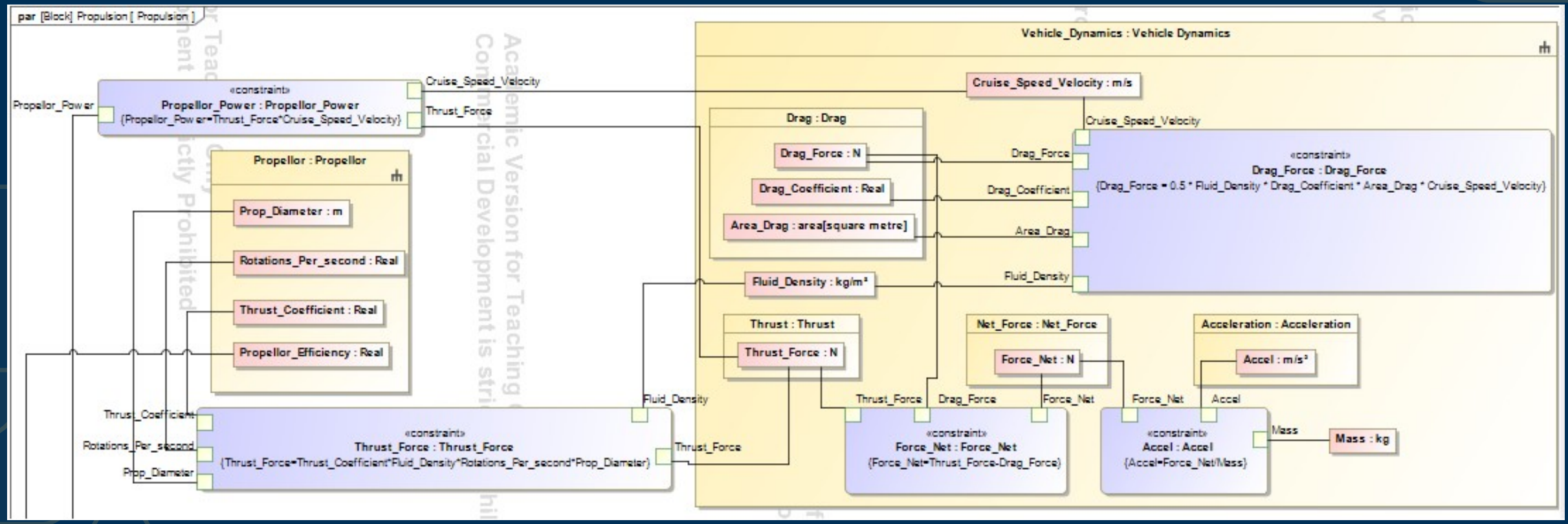
- Drag: $F_D = 0.5\rho C_D A v^2$
  - F_D = Drag force
  - ρ = Fluid density (constant ≈1050)
  - C_D = Drag coefficient (constant ≈ 0.12)
  - A = Reference area (constant ≈ 0.010)
  - v = Velocity (input)
- Thrust: $T = C_T \rho n^2 D^4$
  - T = Thrust force
  - C_T = Thrust coefficient (constant ≈ 0.18)
  - ρ = Fluid density (constant ≈1050)
  - n = Propeller RPM (input)
  - D = Propeller diameter (constant ≈ 0.05)
- Motor Power: $P = \tau\omega$
  - τ = Torque (input)
  - ω = Angular velocity (input)
- Propellor Power: P = Tv
  - P = Power
  - T = Thrust
  - v = Velocity (input)
- Battery Power Needed: $P = P_{prop}/(\eta_{motor} \cdot \eta_{prop})$
  - P = Battery power
  - P_{prop} = Mechanical power
  - η_{motor} = Motor efficiency (constant ≈ 0.95)
  - η_{prop} = Propellor efficiency (constant ≈ 0.80)
- Battery Current: I = P/V
  - I = Battery current
  - P = Battery power
  - v = Battery voltage (constant ≈ Power subsystem)
- Net force: $F_{net} = T - F_D$
  - Fnet = Net force
  - T = Thrust
  - F_D = Drag force
- Acceleration: a = F/m
  - a = Acceleration
  - F = Net force
  - m = Mass (constant ≈ 2.3)

Propulsion BDD Diagram



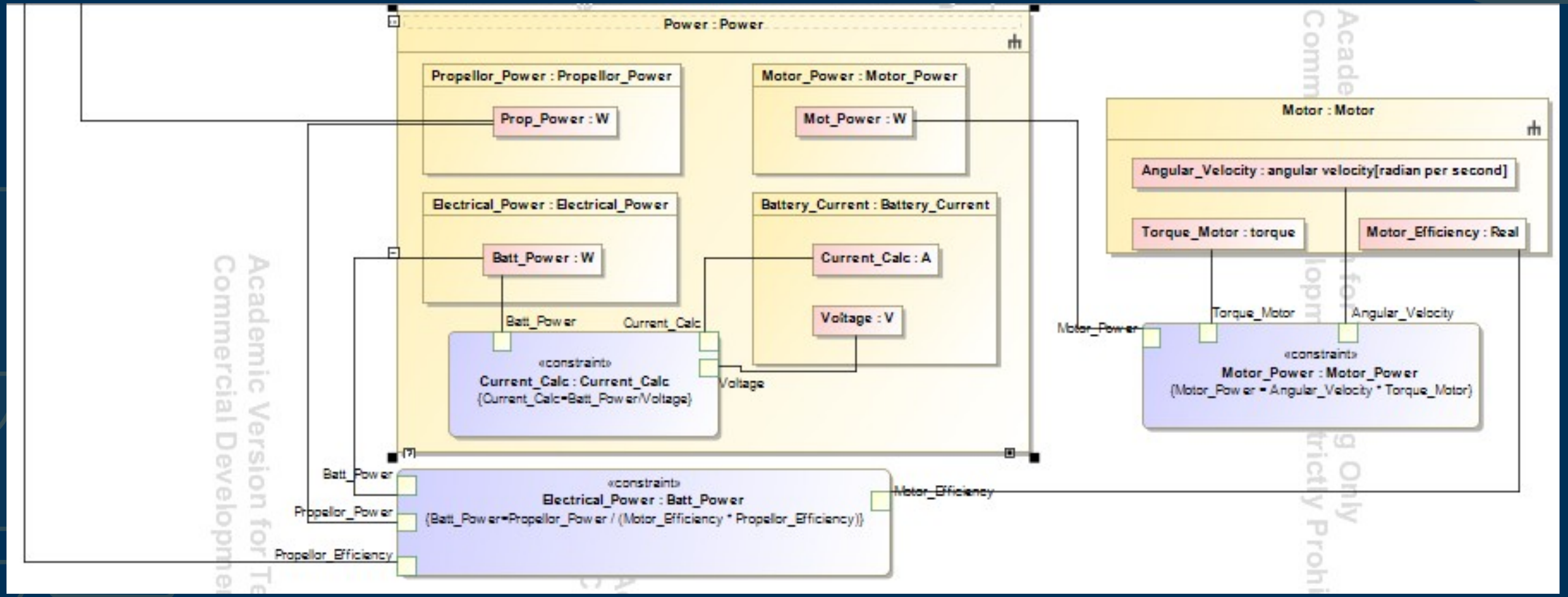
- Objective: Using cruise speed (or a set velocity), determine how much power needed given submarine structure and surrounding forces
- Velocity>>Drag>>Thrust >>(Net force, acceleration)
- Motor>>Propellor Power>>Power Equations

Propulsion Parametric Diagram



- Assumptions:
  - Input: velocity (or cruise speed)
  - No pressure effects from deeper water
  - Comparable to Earth's submarines

Propulsion Parametric Diagram



- Assumptions:
  - Sensors taking angular velocity input
  - No buoyancy effects in this model

Propulsion Simulation Example Results

Thrust_Force : Thrust_Force {Vehicle_Dynamics.Thrust.Thrust_Force=Propellor.T...	propulsion.thrust_Force : Thrust_Force@6f55ec76
Fluid_Density : kg/m³	1050.0000
Prop_Diameter : m	0.0500
Rotations_Per_second : Real	60.0000
Thrust_Coefficient : Real	0.1800
Thrust_Force : N	567.0000

Vehicle_Dynamics : Vehicle Dynamics	propulsion.vehicle_Dynamics : Vehicle Dynamics@69e21e1c
Cruise_Speed_Velocity : m/s	0.2500
Fluid_Density : kg/m³	1050.0000
Mass : kg	2.3000
Acceleration : Acceleration	propulsion.vehicle_Dynamics.acceleration : Acceleration@9f16a84
Accel : m/s²	246.4533
Cruise_Speed : Power	propulsion.vehicle_Dynamics.cruise_Speed : Power@1e32c3b2
Drag : Drag	propulsion.vehicle_Dynamics.drag : Drag@34bec1a6

Drag : Drag	propulsion.vehicle_Dynamics.drag : Drag@34bec1a6
Area_Drag : area[square metre]	0.0100
Drag_Coefficient : Real	0.1200
Drag_Force : N	0.1575

Net_Force : Net_Force	propulsion.vehicle_Dynamics.net_Force : Net_Force@5017abd7
Force_Net : N	566.8425

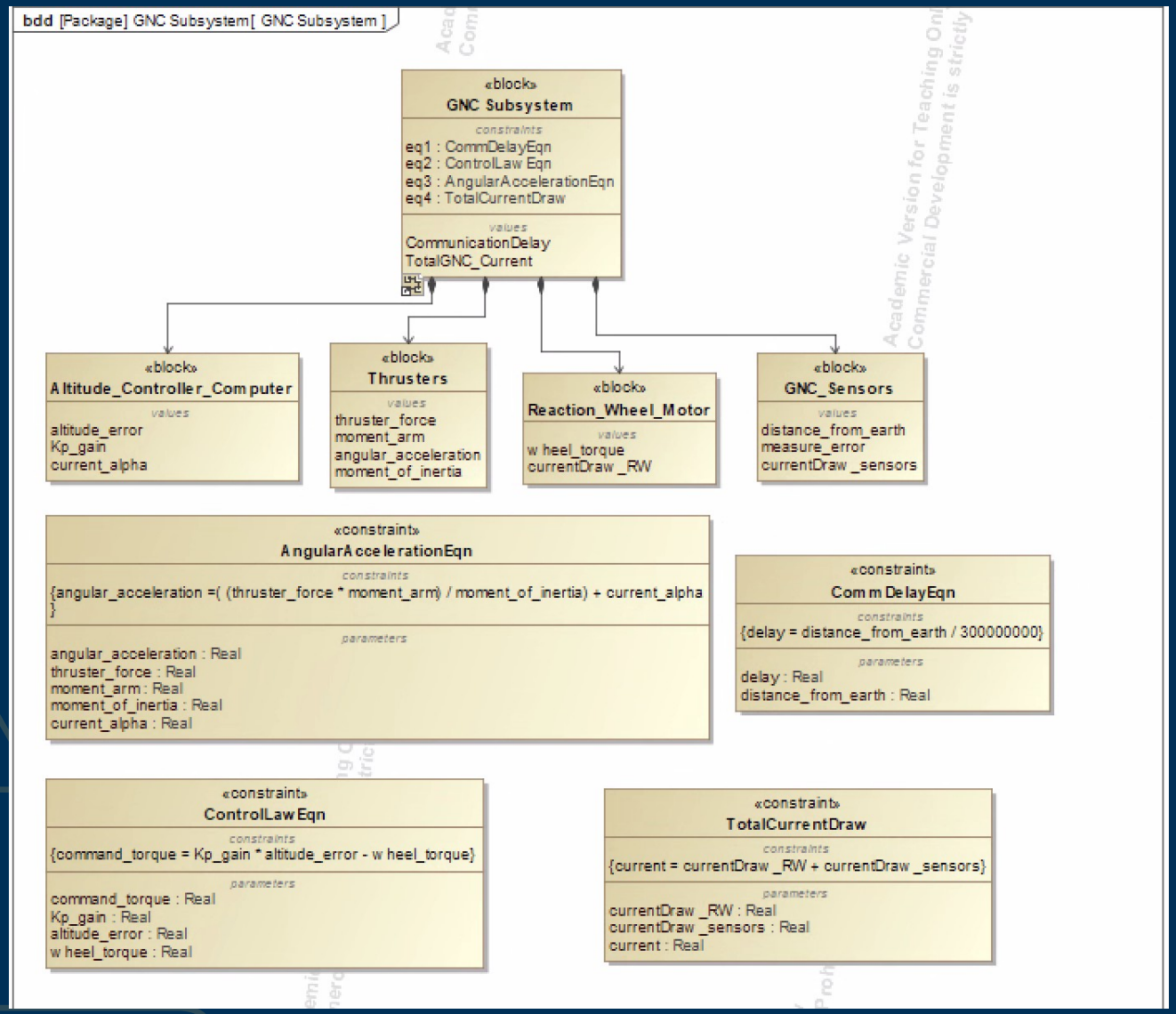
GNC Subsystem

Toby Reitman



Aerospace Systems Design Laboratory

GNC System Architecture



Exploring often used components of control systems:

- Controller computers (Coding)
- Thrusters (TVC)
- Reaction Wheel Motor (internal control)
- GNC Sensors (Relay current state of the system)

Overall BDD contains some of the values that can be obtained using GNC values

Equations Used

Name	CommDelayEqn
Qualified Name	Model::GNC Subsystem::CommDelayEqn::CommDelayEqn
Specification	delay = distance_from_earth / 300000000

Name	AngularAcceleration
Qualified Name	Model::GNC Subsystem::AngularAccelerationEqn::Angular
Specification	angular_acceleration =((thruster_force * moment_arm) / moment_of_inertia) + current_alpha

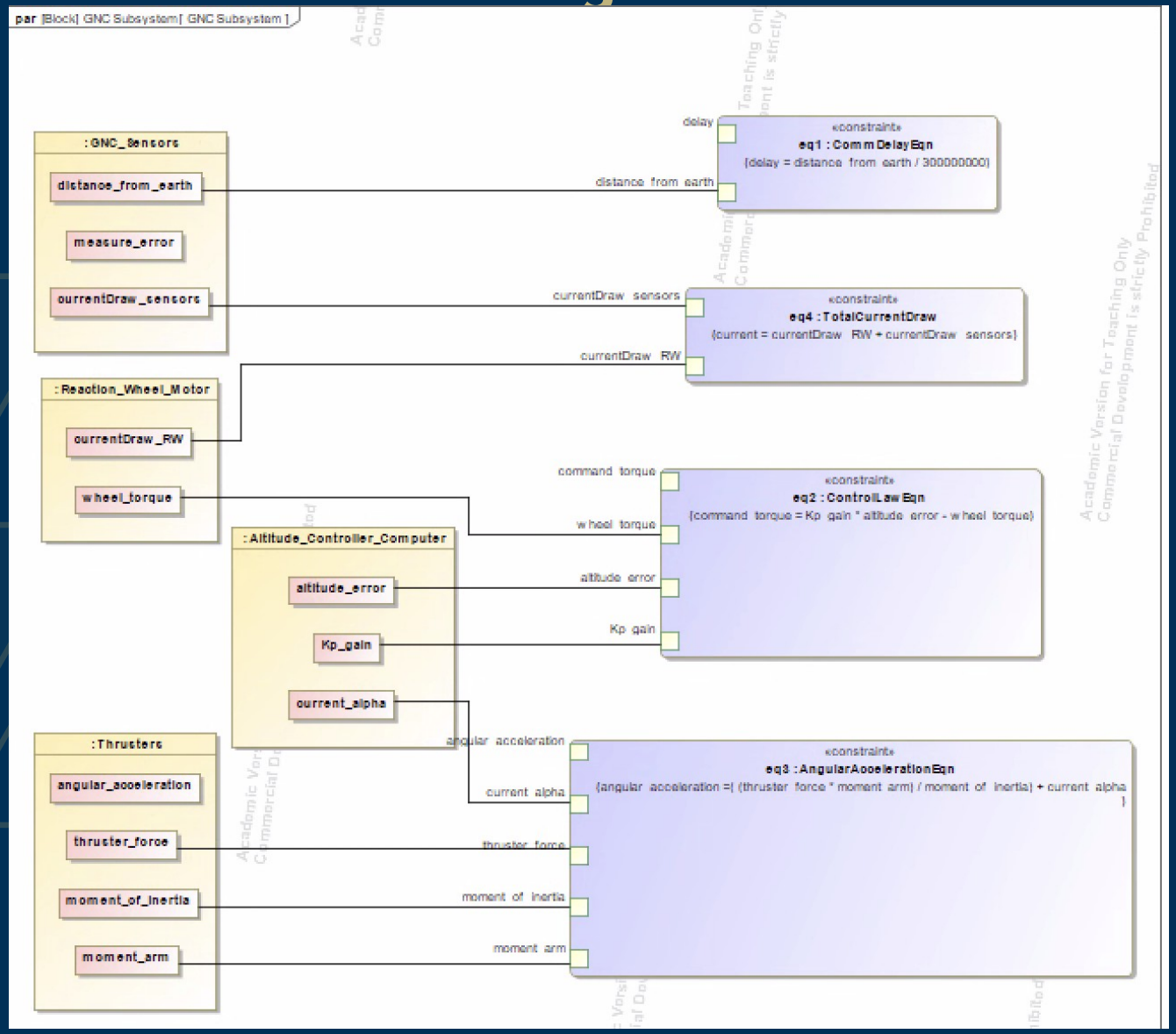
With the given variables from the GNC component , these are some of the equations that could be used to define the system/ could be useful pieces of information

Name	ControlLawEqn
Qualified Name	Model::GNC Subsystem::ControlLawEqn::ControlLawEqn
Specification	command_torque = Kp_gain * altitude_error - wheel_torque

Assume all values used are in SI units

Name	eq4
Qualified Name	Model::GNC Subsystem::TotalCurrentDraw::eq4
Specification	current = currentDraw_RW + currentDraw_sensors

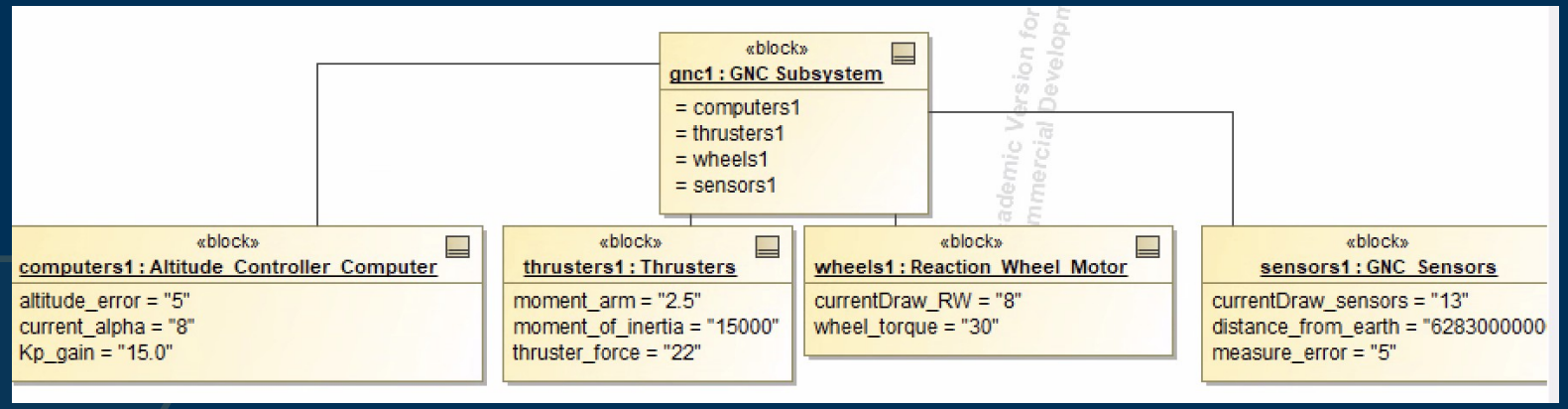
Parametric Diagram



Parametric diagram of sample equations used along with variables

All GNC sub systems depend on one another to solve equations

Sample values used for calculation



Upon running calculations, the model correctly calculated desired values. In reality, the given values would be known quantities or measured quantities by the systems being used

eq1 : CommDelayEqn {delay = .dist...	CommDelayEqn@4a39e5a7
delay : Real	2094.3333
distance_from_earth : Real	6.2830E11

eq2 : ControlLawEqn {command_tor...	ControlLawEqn@54384f61
altitude_error : Real	5.0000
command_torque : Real	45.0000
Kp_gain : Real	15.0000
wheel_torque : Real	30.0000

eq3 : AngularAccelerationEqn {angu...	AngularAccelerationEqn@4c02ce96
angular_acceleration : Real	8.0037
current_alpha : Real	8.0000
moment_arm : Real	2.5000
moment_of_inertia : Real	15000.0000
thruster_force : Real	22.0000

eq4 : TotalCurrentDraw {current = ...	TotalCurrentDraw@32074541
current : Real	21.0000
currentDraw_RW : Real	8.0000
currentDraw_sensors : Real	13.0000



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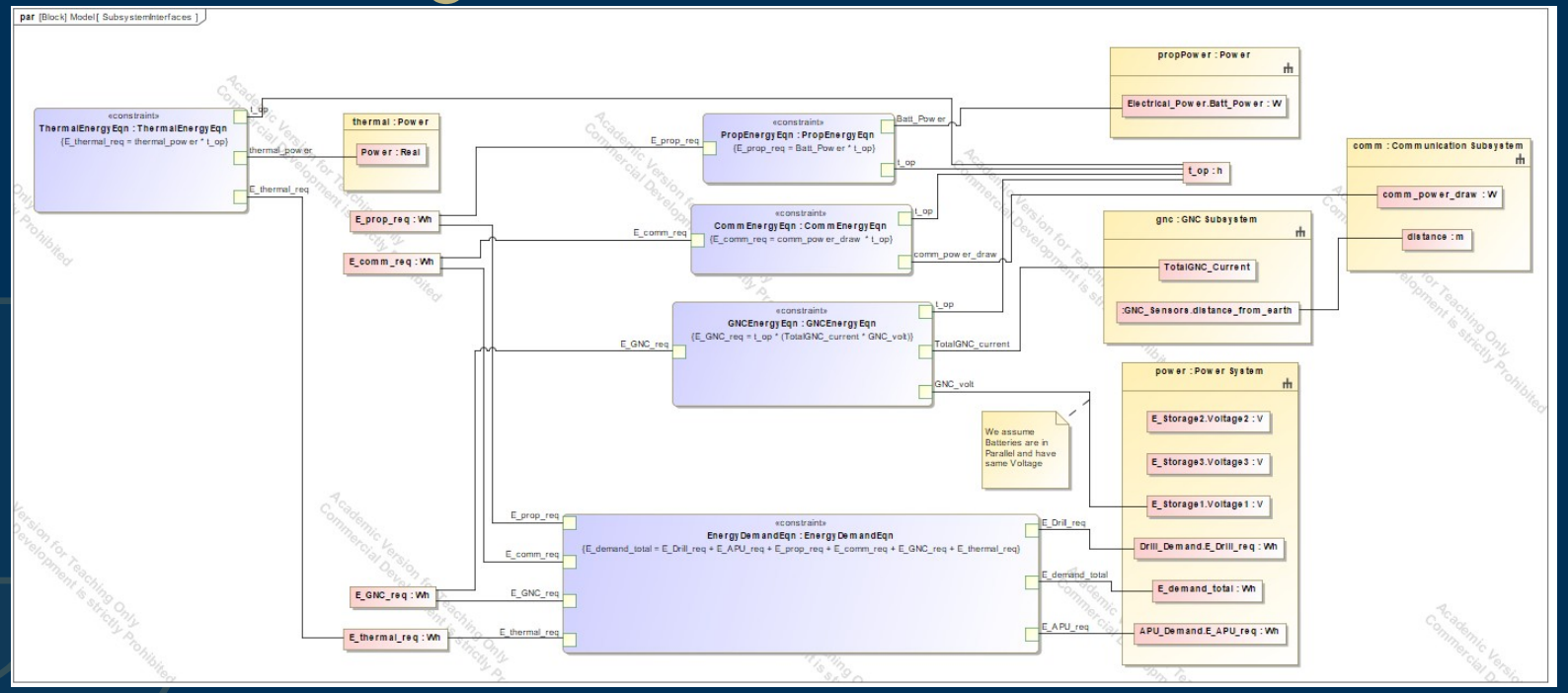
Integrated Subsystems

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Aerospace Systems Design Laboratory

Integration Architecture



- <u>OBJECTIVES:</u>

Connect six independently developed subsystems into one executable model

Establish a common unit so subsystem outputs can be combined

Verify the vehicle's power supply meets total demand

- <u>Approach:</u>

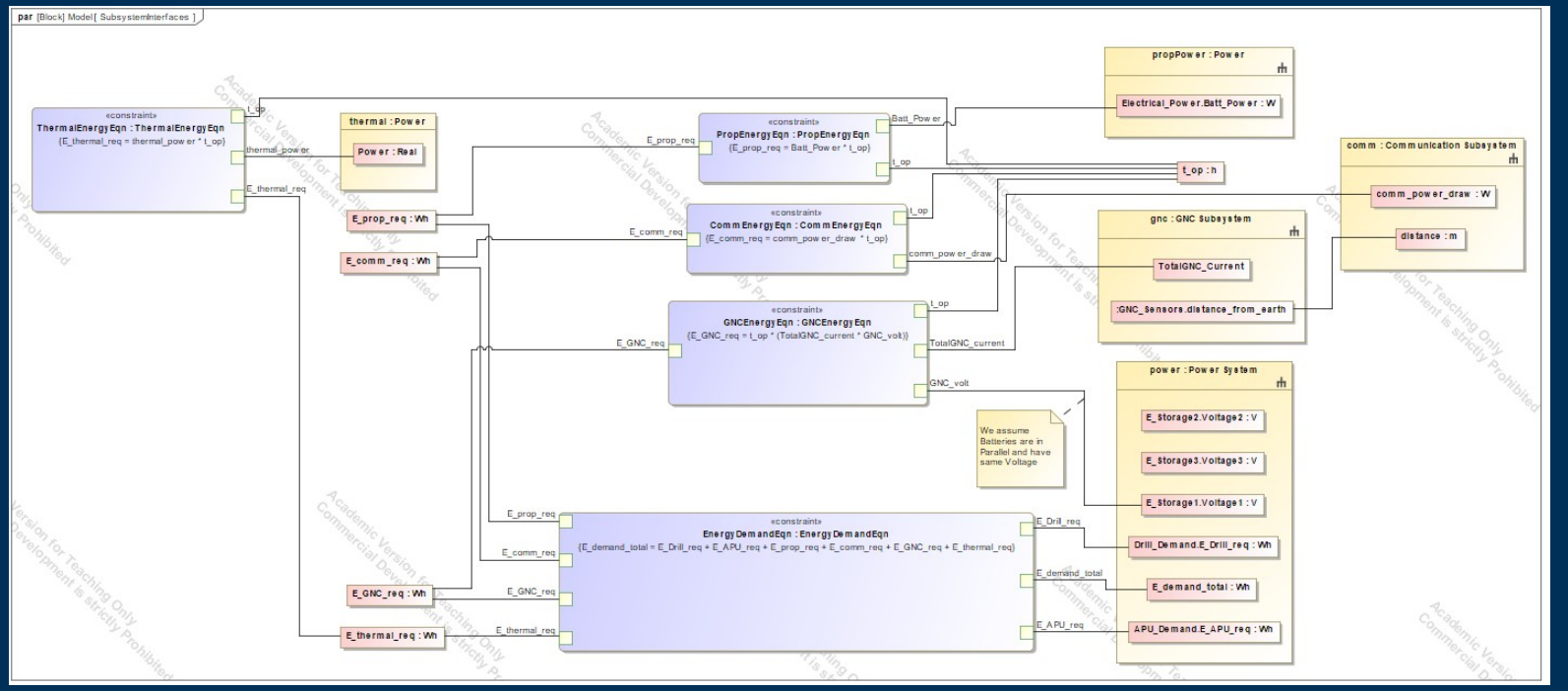
Each Subsystem was modeled independently, then added as a part property of the system level module <block>

A parametric diagram at system level can only reach values in the part properties, hence cross system is possible

Electrical power was chosen as Integrating quantity between almost all subsystems

Additional distance from earth was also connected between systems

Integration Architecture



- Integration Strategy:

Each subsystem's power/amp is converted into energy (Wh) with a given time and battery voltage

Uses assumption that batteries are in parallel and have same Voltage

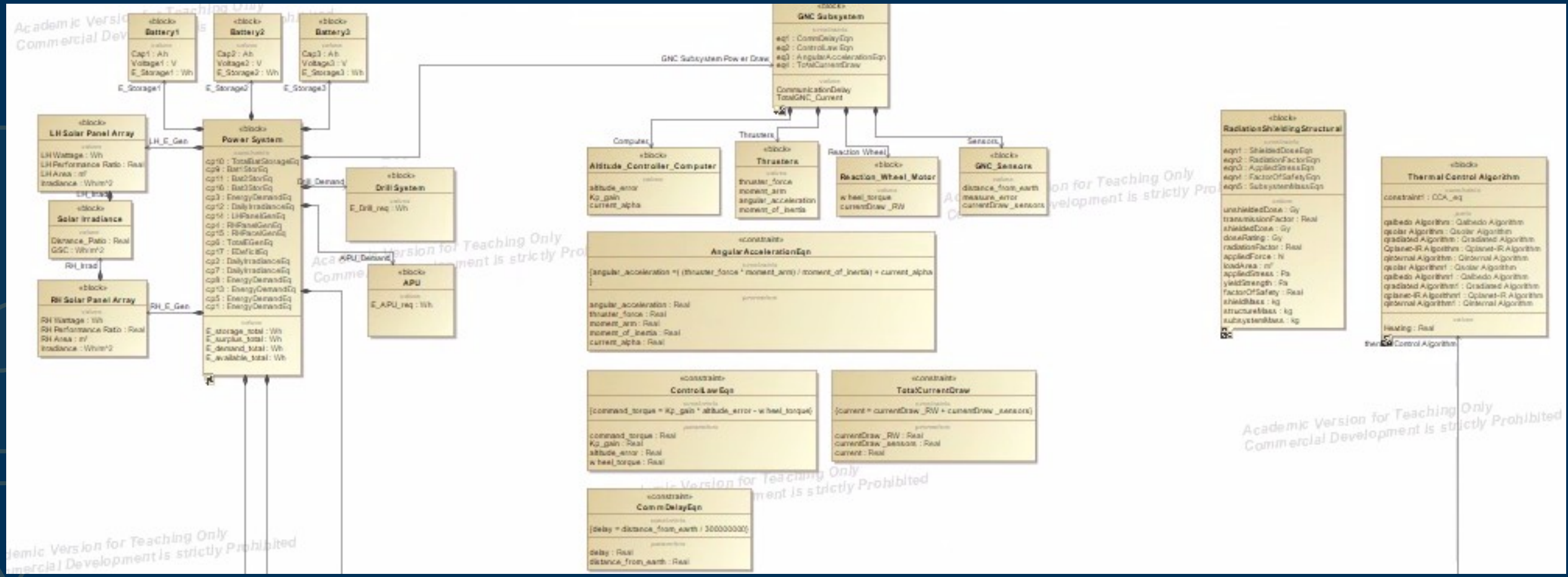
Total Energy is calculated and sent back into power subsystem

Earth to craft distance was connected between GNC and Communications

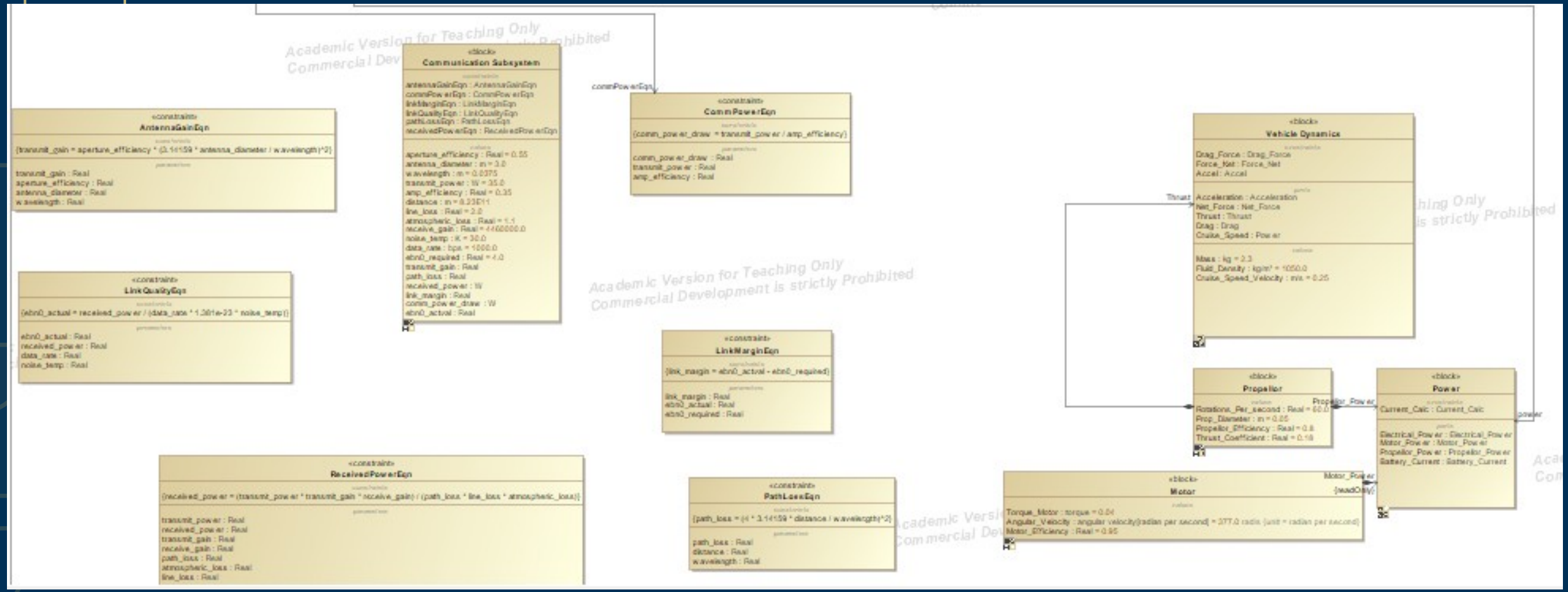
Equations Used

- Comm Energy Demand: $E_{comm_req} = comm_power_draw \cdot t_{op}$
  - E_{comm_req} = daily energy required by the communication subsystem (Wh)
  - $comm_power_draw$ = electrical power drawn by the radio (W)
  - t_{op} = shared operating period (h)
- Propulsion Energy Demand: $E_{prop_req} = Batt_Power \cdot t_{op}$
  - E_{prop_req} = daily energy required by propulsion (Wh)
  - $Batt_Power$ = electrical power drawn from the battery (W)
- GNC Energy Demand: $E_{GNC_req} = t_{op} \cdot (TotalGNC_current \cdot GNC_volt)$
  - E_{GNC_req} = daily energy required by GNC (Wh)
  - $TotalGNC_current$ = summed current draw of reaction wheel and sensors (A)
  - GNC_volt = bus voltage supplied by the batteries (V)
- Thermal Energy Demand: $E_{thermal_req} = thermal_power \cdot t_{op}$
  - $E_{thermal_req}$ = daily energy required by thermal control (Wh)
  - $thermal_power$ = heater power needed to hold operating temperature (W)
- Total Energy Demand: $E_{demand_total} = E_{Drill_req} + E_{APU_req} + E_{prop_req} + E_{comm_req} + E_{GNC_req} + E_{thermal_req}$
  - E_{demand_total} = total daily energy demand of the vehicle (Wh)
  - E_{Drill_req} = energy required by the ice drill (Wh)
  - E_{APU_req} = energy required by the auxiliary power unit (Wh)

Integration BDD



Integration BDD



Academic Version for Teaching Only
Commercial Development is strictly Prohibited

Integration Parametric

